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~~Prims~~

⇒ Algorithm Prims (G)

// purpose : to construct MST

// input : a weighted connected graph

// output : the set of edges which form the MST.

$V_T \leftarrow \{v_0\}$, $E_T \leftarrow \phi$

for $i \leftarrow 1$ to $|V|-1$ do

find an edge $e^* = (v^*, u^*)$ with minimum weight among all edges $e = (v, u)$ such that $v^* \in V_T$ & $u^* \in V - V_T$

$V_T \leftarrow V_T \cup \{u^*\}$

$E_T \leftarrow E_T \cup \{e^*\}$

end for

return E_T

T.C = $O(|E| \log |V|)$

input :

graph [u][v] = {

{0, 2, 0, 6, 0},

{2, 0, 3, 8, 5},

{0, 3, 0, 0, 1},

{4, 5, 0, 0, 9},

{0, 5, 7, 9, 0}}

output :

edge	weight
0 - 1	2
1 - 2	3
0 - 3	6
1 - 4	5